

Project 2: Speech Enhancement

EE4715: Array Processing
Delft University of Technology

Burcu Erarslanoglu (5948355) Joris van de Weg (5174821)

1 Introduction

Speech enhancement is a critical area of research and application that aims to improve the intelligibility and quality of speech signals. Digital speech communication systems are designed to work with noise-free speech; however, due to non-ideal channels, echo, competing speakers, or other types of noises the speech signals may get degraded. Because of this, speech enhancement techniques are vital for various applications including mobile communication systems, hearing aids, and voice-controlled systems. This project focuses on designing, implementing, and evaluating algorithms for speech enhancement using a multi-microphone system in MATLAB. The main goal is to develop a beamforming system for far-end noise reduction which essentially means reducing the effect of unwanted noise that originates from a remote location. Both a single microphone and a multi-microphone array structure can be used in a far-end noise reduction setting. As sound signals $s[n]$ propagate through a medium, they get affected by the properties of that medium. To capture the reflections, reverberations, and echoes in that medium, a room impulse response $h[n]$ is used, resulting in the following signal model:

$$x[n] = (h * s)[n]. \quad (1)$$

This single microphone model can be extended to the case where there are d point sources s_i and the room impulse response h_i for each of the sources from source position to the microphone differs. In addition, noise n can be added to the signal model resulting in:

$$x[n] = \sum_{i=1}^d (h_i * s_i)[n] + n[n]. \quad (2)$$

To analyze the signal in frequency domain, the Short-Time Fourier Transform (STFT) is applied to the time-domain signal $x[n]$. As a result of the transformation $X(k, l) \leftrightarrow \text{STFT}\{x[n]\}$, time-domain signal's time-frequency representation is achieved where k represents the frequency bin and l represents the time frame. With this transformation and extension of the single microphone model to multiple microphone model using stacked vector notation eq. (2) is written as:

$$\mathbf{x}(k, l) = \sum_{i=1}^d a_i(k, l) s_i(k, l) + \mathbf{n}(k, l), \quad (3)$$

where $\mathbf{x}(k, l)$ is the stacked vector of signals received by M microphones, $a_i(k, l)$ is the complex gain (including the effect of beamforming) for the i -th source at the k -th frequency bin and l -th time frame, $s_i(k, l)$ remains the source signal in the frequency domain, and $\mathbf{n}(k, l)$ is the noise vector in the frequency domain. The “short-time” in STFT comes from the method's approach of analyzing a signal over short, overlapping “windows” of time. By doing so, the signal's frequency content as they change over time can be captured in a better way. The shape of the windowing is also relevant. If a regular rectangular window is applied, the sharp edges cause spectral leakage; therefore, typically window functions that are centered on the window segment and tapered off towards the edges are used. This allows for smooth transition at the boundaries, and reduces spectral leakage.

In this project's problem setup, five different sound signals will be used and using these far-end noise reduction is aimed. Among these, two are clean speech signals. These sound signals will be combined to create mixtures on which speech enhancement methods will be applied to estimate the intended speech. The setup includes $M = 4$ microphones and $d = 5$ sound sources. Each source

has a unique impulse response h_i for each microphone, which characterizes the reflections. The dataset using in the project including the speech signals and microphone to sound source impulse responses are given in Table 1.

Table 1: Datasets for the project.

(a) Speech Signals		(b) Multi-microphone Impulse Responses	
Speech Signal	Duration [sec]	Impulse Response	Size
Clean Speech 1	36	h_inter1	4×400
Clean Speech 2	33	h_inter2	4×400
Artificial Non-Stationary Noise	36	h_inter3	4×400
Stationary Speech Shaped Noise	90	h_inter4	4×400
Babble Noise	37	h_target	4×400

The STFT is applied on the mixed sound signals by using a 50% overlapping Hanning Window. Finally, the project examines six different scenarios: a mixture of Clean Speech 1 with Artificial Non-Stationary Noise, a mixture of Clean Speech 1 with Stationary Speech Shaped Noise, a mixture of Clean Speech 1 with Babble Noise, a mixture of Clean Speech 1 with all the noise types, a mixture of Clean Speech 1 with Clean Speech 2, a mixture of Clean Speech 1 with all the noise types and Clean Speech 2.

2 Acoustic Transfer Function

The acoustic transfer function (ATF) describes the relationship between the signal sources and the received signal at the microphone. This relationship is affected by the distance the signal travels and the environmental and medium conditions through which the signal propagates. The ATF, needs to be derived to define multiple beamformers, which will be discussed in the following section. We are especially interested in the frequency domain representation of the ATF, $a(k, l)$. Recall, the earlier-mentioned signal model:

$$x_m(k, l) = a_m(k, l)s(k, l) + n_m(k, l). \quad (4)$$

To simplify the notation, we can express the signals in vector form. Let $\mathbf{x}(k, l)$, $\mathbf{a}(k, l)$, and $\mathbf{n}(k, l)$ represent the received signal, ATF, and noise vectors across all microphones, respectively. Thus, the received signal vector can be written as:

$$\mathbf{x}(k, l) = \mathbf{a}(k, l)s(k, l) + \mathbf{n}(k, l). \quad (5)$$

The ATF can be estimated using the covariance matrices of the received signals. The covariance matrix of the received signal $\mathbf{R}_x(k, l)$ is given by:

$$\mathbf{R}_x(k, l) = \mathbb{E}[\mathbf{x}(k, l)\mathbf{x}^H(k, l)]. \quad (6)$$

Similarly, the covariance matrix of the noise $\mathbf{R}_n(k, l)$ is:

$$\mathbf{R}_n(k, l) = \mathbb{E}[\mathbf{n}(k, l)\mathbf{n}^H(k, l)]. \quad (7)$$

In the case the noise is spatially white, the noise covariance matrix can be approximated as:

$$\mathbf{R}_n(k, l) = \sigma_n^2 \mathbf{I}, \quad (8)$$

where σ_n^2 is the variance of the noise, which is the same as noise power. Due to the spatially white noise, the noise is uncorrelated over the microphones, and therefore the variance is the same for all microphones.

For this paper, there is one target signal, s_{target} , which is our signal of interest. This means that all the other signals recorded by the microphones are interference and noise:

$$\mathbf{x}(k, l) = \mathbf{a}_{\text{target}}(k, l)s_{\text{target}}(k, l) + \underbrace{\sum_{i=2}^d \mathbf{a}_i(k, l)s_i(k, l)}_{\text{interferers} + \text{noise}} + \mathbf{n}'(k, l). \quad (9)$$

Assuming the signal of interest and the other unwanted signal sources are uncorrelated, the covariance of the measured data can be written as

$$\mathbf{R}_x = \mathbf{R}_s + \mathbf{R}_n. \quad (10)$$

The auto correlation matrix for the target signal, $\mathbf{R}_s$, is defined as:

$$\mathbf{R}_s = \mathbb{E}[s(k, l)s(k, l)^H] = \sigma_{s_1}^2 \mathbf{a}(k, l)\mathbf{a}^H(k, l), \quad (11)$$

where $s(k, l) = \mathbf{a}(k, l)s_{\text{target}}(k, l)$ and $\sigma_{s_1}^2 = \mathbb{E}[s_1(k, l)(s_1(k, l))^H]$. One important property is because there is only one source $\text{rank}(\mathbf{R}_s) = 1$. As Equation 11 shows, the ATF can be derived by finding $\mathbf{R}_s$ and then applying the eigenvalue decomposition.

2.1 Estimation Using Eigenvalue Decomposition

Usually, the noise and other interference sources are not spatially white noises and thus $\mathbf{R}_n$ cannot be written as in equation 8. Pre-whitening is a technique which solves this problem. It can transform the received signal such that the correlated noise process becomes a white noise process. The first step of pre-whitening comes with computing the Hermitian square root and its inverse of the noise covariance matrix $\mathbf{R}_n$. The eigendecomposition can be written as:

$$\mathbf{R}_n = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H, \quad (12)$$

where $\mathbf{U}$ is the unitary matrix of eigenvectors and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues. $\mathbf{R}_n$ is the covariance matrix, which is Hermitian and positive semi-definite. The Hermitian square root of $\mathbf{R}_n$ is given by:

$$\mathbf{R}_n^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^H. \quad (13)$$

The inverse of the Hermitian square root of $\mathbf{R}_n$ is:

$$\mathbf{R}_n^{-1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^H. \quad (14)$$

The observed data is then transformed to contain white noise, using the $\mathbf{R}_n^{-1/2}$:

$$\tilde{\mathbf{x}}(k, l) = \mathbf{R}_n^{-1/2}\mathbf{x}(k, l). \quad (15)$$

The autocorrelation matrix $\mathbf{R}_x$ is also transformed:

$$\tilde{\mathbf{R}}_x(k, l) = \mathbf{R}_n^{-1/2}\mathbf{R}_x(k, l)\mathbf{R}_n^{-1/2}. \quad (16)$$

With these whitened matrices, the process of retrieving the $\mathbf{R}_x(k, l)$ will become possible. Using Equations 11 and 8, $\tilde{\mathbf{R}}_x$ can, by decomposing the Hermitian PSD matrix $\tilde{\mathbf{R}}_s$, be rewritten as follows:

$$\begin{aligned}\tilde{\mathbf{R}}_x &= \tilde{\mathbf{R}}_s + \sigma_n^2 \mathbf{I} \\ &= \tilde{\mathbf{U}} \mathbf{\Lambda} \tilde{\mathbf{U}}^H + \sigma_n^2 \mathbf{I} \\ &= \tilde{\mathbf{U}} \mathbf{\Lambda} \tilde{\mathbf{U}}^H + \sigma_n^2 \tilde{\mathbf{U}} \tilde{\mathbf{U}}^H \\ &= \tilde{\mathbf{U}} (\mathbf{\Lambda} + \sigma_n^2 \mathbf{I}) \tilde{\mathbf{U}}^H,\end{aligned}\tag{17}$$

where $\tilde{\mathbf{U}}$ is the unitary matrix. Following, only $\mathbf{U}_1$ is kept, referring to the unitary matrix associated with the largest eigenvalue. The signal covariance matrix from the found principal subspace can be derived as:

$$\hat{\mathbf{R}}_s = \mathbf{U}_1 \mathbf{\Lambda}_1 \mathbf{U}_1^H.\tag{18}$$

Finally, the estimated $\hat{\mathbf{R}}_s$ can be transformed back to the original space. To obtain the signal covariance matrix in the context of the original data, the $\hat{\mathbf{R}}_s$ is de-whitened:

$$\hat{\mathbf{R}}_s = \mathbf{R}_n^{1/2} \tilde{\mathbf{U}}_1 \mathbf{\Lambda}_1 \tilde{\mathbf{U}}_1^H \mathbf{R}_n^{1/2}.\tag{19}$$

This estimated covariance signal matrix can be decomposed,

$$\hat{\mathbf{R}}_s = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H\tag{20}$$

where the ATF $\mathbf{a}(k, l)$ can be approximated by the eigenvector corresponding to the largest eigenvalue, due to having one target source.

2.2 Estimation Using Generalized Eigenvalue Decomposition

As shown in the previous subsection, pre-whitening demands computing the negative square root of the noise covariance matrix $\mathbf{R}_n^{1/2}$. This computation can result in a loss of accuracy in the data due to the non-trivial operation. This can be avoided by working directly with $\mathbf{R}_x$ and $\mathbf{R}_n$. Such an approach for estimating the signal covariance matrix $\mathbf{R}_s$ is the Generalized Eigenvalue Decomposition (GEVD).

Using GEVD, the goal is to decompose the observed signal covariance matrix $\mathbf{R}_x$ (Equation 10) to isolate the signal component $\mathbf{R}_s$. Given the Hermitian matrices $\mathbf{A}$ and $\mathbf{B}$ with $\mathbf{B} \succ 0$, there exists a non-singular matrix $\mathbf{U}$, such that:

$$\mathbf{U}^H \mathbf{A} \mathbf{U} = \mathbf{\Lambda}_A \quad \text{and} \quad \mathbf{U}^H \mathbf{B} \mathbf{U} = \mathbf{\Lambda}_B,\tag{21}$$

where $\mathbf{\Lambda}_A$ and $\mathbf{\Lambda}_B$ are diagonal matrices. Hence, the following can be derived:

$$\mathbf{B} \mathbf{U} = \mathbf{U}^{-H} \mathbf{\Lambda}_B \mathbf{U}^H \mathbf{U} = \mathbf{U}^{-H} \mathbf{\Lambda}_B \mathbf{U}^H \quad \text{and} \quad \mathbf{A} \mathbf{U} = \mathbf{U}^{-H} \mathbf{\Lambda}_A = \mathbf{U}^{-H} \mathbf{\Lambda}_A \mathbf{U} \mathbf{\Lambda}_B^{-1} \mathbf{\Lambda}_A \mathbf{U}^H \mathbf{U} \mathbf{\Lambda},\tag{22}$$

which gives:

$$\mathbf{A} \mathbf{U} = \mathbf{B} \mathbf{U} \mathbf{\Lambda} \quad \text{where} \quad \mathbf{\Lambda} = \text{diag}\left(\frac{\alpha_1}{\beta_1}, \dots, \frac{\alpha_n}{\beta_n}\right)\tag{23}$$

This decomposition is known as the generalized eigenvalue decomposition (GEVD). Recalling that $\mathbf{B}$ is a positive definite matrix and therefore invertible, the expression can be rewritten as:

$$\mathbf{B}^{-1} \mathbf{A} \mathbf{U} = \mathbf{U} \mathbf{\Lambda} \quad \text{where} \quad \mathbf{\Lambda} = \text{diag}\left(\frac{\alpha_1}{\beta_1}, \dots, \frac{\alpha_n}{\beta_n}\right)\tag{24}$$

Applying this to the signal and noise covariance matrices $\mathbf{R}_s$ and $\mathbf{R}_n$, and setting $\beta_i = 1$ for all i , we have:

$$\mathbf{U}^H \mathbf{R}_s \mathbf{U} = \mathbf{\Lambda} \quad \text{and} \quad \mathbf{U}^H \mathbf{R}_n \mathbf{U} = \mathbf{I}_M, \quad (25)$$

where $\mathbf{\Lambda} \succeq 0$. Hence, the pair $(\mathbf{\Lambda}, \mathbf{U})$ are the eigenvalues/vectors of the matrix $\mathbf{R}_n^{-1} \mathbf{R}_s$, and $\mathbf{Q}$ are the left eigenvectors of $\mathbf{R}_n^{-1} \mathbf{R}_s$. From equation 10, the following can be derived.

$$\mathbf{U}^H \mathbf{R}_x \mathbf{U} = \mathbf{\Lambda} + \mathbf{I}_M \quad \Longleftrightarrow \quad \mathbf{R}_x = \mathbf{U}^{-H} (\mathbf{\Lambda} + \mathbf{I}_M) \mathbf{U}^{-1}. \quad (26)$$

When the rank of $\mathbf{R}_s$, r , is smaller than number of microphones, $\mathbf{R}_x$ can be partitioned as:

$$\mathbf{R}_x = (\mathbf{Q}_1 \quad \mathbf{Q}_2) \begin{pmatrix} \mathbf{\Lambda}_1 + \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_{M-r} \end{pmatrix} \begin{pmatrix} \mathbf{Q}_1^H \\ \mathbf{Q}_2^H \end{pmatrix}, \quad (27)$$

Similarly to the previous section, the estimate of $\mathbf{R}_s$ can be derived from the found the GEVD of $\mathbf{R}_x$ as:

$$\hat{\mathbf{R}}_s = \mathbf{Q}_1 (\mathbf{\Lambda}_1 + \mathbf{I}_r) \mathbf{Q}_1^H, \quad (28)$$

or, by reducing the remaining eigenvalues by one:

$$\hat{\mathbf{R}}_s = \mathbf{Q}_1 \mathbf{\Lambda}_1 \mathbf{Q}_1^H. \quad (29)$$

Because for this paper's problem, only one target source is present as seen in Table 1, $\text{rank}(\mathbf{R}_s(k, l)) = 1$ thus the ATF can be obtained by selecting the $\mathbf{Q}_1$, the principal generalized eigenvector between $\mathbf{R}_s(k, l)$ and $\mathbf{R}_n(k, l)$.

3 Beamformers

Beamforming is a technique in signal processing to enhance the signal coming from a particular direction while suppressing the signals coming from other directions. The main goal is to improve the quality of the received signal which can be measured with the metric of output signal-to-noise (SNR) ratio. Due to its advantages, beamforming and beamformers are used in various domains including audio signal processing, radar, and wireless communication applications. For beamforming, it is assumed that the sound signal is narrowband. This simplifies the process because it allows the use of fixed phase shifts and weights to combine the signals from different microphones. This approach works because the phase shift introduced by the propagation delay becomes approximately the same across the narrow frequency range that the signal is in. Although speech signals are traditionally not narrowband, they can be treated as narrowband over short time intervals for practical purposes. This assumption simplifies many speech processing tasks, including beamforming and application of the STFT discussed in section 1. There are different types of beamformers, each with their own approach and the ones that are used in this project are discussed in this section in detail.

3.1 Delay and Sum (DAS) Beamformer

The general idea of the Delay and Sum (DAS) Beamformer is the coherent summation of the signals received at each microphone in the array after applying a complementary time delay to each signal. If the signal is assumed to be narrowband, time delays become a phase shift in the frequency domain and they are then implemented by the multiplication of the microphone signals with complementary phase shifts. In the setup for the DAS Beamformer, free-field environment is

assumed and the signals are narrowband. When the first microphone is selected as the reference point, the observed signal is modeled as follows:

$$\mathbf{x}(k, l) = s_1(k, l)\mathbf{a}(k, l) + \mathbf{n}(k, l). \quad (30)$$

In the eq. (30), $\mathbf{x}(k, l)$ is the vector of received signals, $s_1(k, l)$ is the source signal, $\mathbf{a}(k, l)$ is the ATF, and $\mathbf{n}(k, l)$ represents noise. Knowing $\mathbf{a}(k, l)$ from the steps given in section 2, the output of the DAS Beamformer can be computed as follows:

$$\hat{s}(k, l) = \mathbf{w}^H(k, l)\mathbf{x}(k, l). \quad (31)$$

The weight vector $\mathbf{w}(k, l)$ is defined in different ways for the near free-field, far free-field and general cases. For a near and free-field scenario, the weight vector is defined as:

$$\mathbf{w}(k, l) = \frac{\mathbf{a}(k, l)}{\mathbf{a}^H(k, l)\mathbf{a}(k, l)}. \quad (32)$$

In the case of a far and free-field, the weight vector simplifies to:

$$\mathbf{w}(k, l) = \frac{1}{M}\mathbf{a}(k, l), \quad (33)$$

where M is the number of microphones. For the most general scenarios (non-free field), the weight vector expression is the same as eq. (32). Because the microphone location and signal direction might be unknown, finding the complementary delays for each microphone might be a challenging task. On the other hand, the DAS beamformer does not require any specific knowledge about the characteristics of $\mathbf{n}(k, l)$ and it preserves the desired signal from the target source. Also, it is able to reduce the noise variance in most ideal case where the noise is uncorrelated across microphones.

3.2 Minimum Variance Distortionless Response (MVDR) Beamformer

The Minimum Variance Distortionless Response (MVDR) Beamformer can be characterized as a more advanced beamformer because it is able to exploit the position of target and noise sources in an adaptive manner. The received signal vector $\mathbf{x}(k, l)$ is taken as it is given in eq. (5). The MVDR puts a constraint on the beamformer to have no change of magnitude and phase in the direction of the target source. With this constraint, also known as the distortionless constraint, the MVDR solves a cost function that minimizes the variance of the beamformer output in all other directions. When the mentioned cost function is solved the weight vector is given as follows.

$$\mathbf{w}(k, l) = \frac{\mathbf{R}_x^{-1}(k, l)\mathbf{a}(k, l)}{\mathbf{a}^H(k, l)\mathbf{R}_x^{-1}(k, l)\mathbf{a}(k, l)} = \frac{\mathbf{R}_n^{-1}(k, l)\mathbf{a}(k, l)}{\mathbf{a}^H(k, l)\mathbf{R}_n^{-1}(k, l)\mathbf{a}(k, l)} \quad (34)$$

In the second part of the expression for $\mathbf{w}(k, l)$, the weight vector is written using the noise correlation matrix $\mathbf{R}_n$. The transformation using matrix inversion lemma is possible by writing the following:

$$\mathbf{R}_x(k, l) = \mathbf{R}_n(k, l) + \mathbf{a}(k, l)\mathbf{a}^H(k, l)\sigma_s^2(k, l). \quad (35)$$

The expression in eq. (35) and consequently the full expression in eq. (34) is true under the three assumptions that signal correlation matrix $\mathbf{R}_s$ is full rank, target and noise are uncorrelated; and target and noise are additive. In the case where the noise field is spatially uncorrelated, $\mathbf{R}_n(k, l) = \sigma_N^2(k, l)\mathbf{I}_M$, the MVDR equals to the DAS beamformer given in eq. (32) and eq. (33). Due to the constraint applied, MVDR is able to preserve the desired signal structure while minimizing the contributions of noise and interference; however due to its adaptive nature, computing the matrix inversion lemma every time when the noise statistics change when computing $\mathbf{w}(k, l)$ is computationally expensive and slow.

3.3 Optimal Linear Multi-Channel Wiener Beamformer

For the Optimal Linear Multi-Channel Wiener Beamformer, again $\mathbf{x}(k, l)$ is taken from eq. (5). By minimizing the mean-square error (MSE) between the estimated signal $\hat{s}(k, l) = \mathbf{w}^H(k, l)\mathbf{x}(k, l)$ and true signal $s(k, l)$, the Optimal Linear Multi-Channel Wiener Beamformer is able to find the weight vector as given in the following:

$$\mathbf{w}(k, l) = \mathbf{R}_x^{-1}(k, l)\sigma_s^2(k, l)\mathbf{a}(k, l). \quad (36)$$

The matrix inversion lemma is used for the Optimal Linear Multi-Channel Wiener Beamformer as well, to transform it into the following structure:

$$\mathbf{w}(k, l) = \frac{\sigma_s^2(k, l)}{\sigma_s^2(k, l) + (\mathbf{a}^H(k, l)\mathbf{R}_n^{-1}(k, l)\mathbf{a}(k, l))^{-1}} \frac{\mathbf{R}_n^{-1}(k, l)\mathbf{a}(k, l)}{\mathbf{a}^H(k, l)\mathbf{R}_n^{-1}(k, l)\mathbf{a}(k, l)}, \quad (37)$$

where the first fraction corresponds to the single-channel Wiener filter and second fraction corresponds to the already introduced MVDR beamformer in eq. (34). The MVDR part of the beamformer is able to suppress noise and interference with spatial filtering while the single-channel Wiener filter post-processes the MVDR filter to further reduce the noise interference.

4 Evaluation Metrics

Different evaluation metrics can be used to understand the performance of a certain speech enhancement technique specifically beamformers. The ones that will be used in this project are output SNR, and short-time objective intelligibility measure (STOI). In this section, these evaluation metrics will be explained.

4.1 Output Signal-to-Noise (SNR) Ratio

The output SNR is calculated by using the weight vector $\mathbf{w}$, source signal vector $\mathbf{s}$, and noise vector $\mathbf{n}$. The calculation is given as follows:

$$\text{SNR}_{\text{out}}(\mathbf{w}) = \frac{\mathbb{E}[|\mathbf{w}^H \mathbf{s}|^2]}{\mathbb{E}[|\mathbf{w}^H \mathbf{n}|^2]} = \frac{\mathbf{w}^H \mathbf{R}_s \mathbf{w}}{\mathbf{w}^H \mathbf{R}_n \mathbf{w}}. \quad (38)$$

The computed SNR_{out} is a real-valued function of the complex vector $\mathbf{w}$. In practice, the SNR_{out} can be maximized by writing a steepest descent formulation. During the evaluations, the weight vector $\mathbf{w}$ will be derived from their corresponding versions of the beamformer forms discussed in section 3.

4.2 Short-Time Objective Intelligibility Measure (STOI)

The Short-Time Objective Intelligibility (STOI) measure is a widely used metric for predicting the intelligibility of speech signals processed by enhancement algorithms [1]. The STOI is a function of clean and degraded speech signals, denoted by x and y , respectively. The output that STOI gives is a scalar value which is monotonic with the average intelligibility of y . The average intelligibility of y refers to the percentage of words correctly understood by a group of listeners. First, signals x and y are time-frequency (TF) decomposed to obtain a simplified internal representation resembling the auditory system's transform properties. This involves segmenting both signals into 50% overlapping, Hann-windowed frames with a length of 256 samples, zero-padded to 512 samples. Silent regions that do not contribute to speech intelligibility are removed by excluding frames where the clean speech energy is lower than 40 dB with respect to the maximum clean speech energy frame. A one-third octave band analysis is performed by grouping DFT-bins, which simplifies

the frequency analysis and make it focus on the parts of the spectrum that are most relevant for speech intelligibility. The energy in the j -th one-third octave band of the clean speech is given by: given below:

$$X_j(m) = \sqrt{\sum_{k=k_1(j)}^{k_2(j)-1} |\hat{x}(k, m)|^2} \quad (39)$$

4.3 One-Third Octave Band Analysis

One-third octave bands are used to simplify the frequency analysis by grouping frequency components into bands. This is done to focus on the parts of the frequency spectrum that are most relevant for speech intelligibility. The energy in the j -th one-third octave band of the clean speech is given by:

$$X_j(m) = \left(\sum_{k=k_l(j)}^{k_u(j)-1} |x(k, m)|^2 \right)^{1/2}, \quad (40)$$

where $x(k, m)$ is the discrete Fourier transform (DFT) of the clean speech signal at frequency bin k and frame m and $k_l(j)$ and $k_u(j)$ are the lower and upper-frequency bin indices.

To capture the dynamics of speech, the temporal envelope is calculated. This is the smoothed representation of the signal's amplitude variations over time. In other words, it tells how the loudness of the speech signal changes, which is important to understand spoken words. The temporal envelope is calculated for each frame both the clean (x_j, m) and degraded ($y_{j,m}$) speech, and is denoted as:

$$\mathbf{x}_{j,m} = [X_j(m - N + 1), X_j(m - N + 2), \dots, X_j(m)]^T \quad (41)$$

. The normalized version of $\mathbf{y}_{j,m}$, denoted by $\bar{\mathbf{y}}_{j,m}$, is given by:

$$\bar{y}_{j,m}(n) = \min \left(\frac{\|\mathbf{x}_{j,m}\|}{\|\mathbf{y}_{j,m}\|} y_{j,m}(n), (1 + 10^{\beta/20}) x_{j,m}(n) \right) \quad (42)$$

where β (dB) refers to the lower signal-to-distortion ratio (SDR) bound. The SDR measures how much the degraded speech signal deviates from the clean speech signal. In other words, a higher SDR indicates that the degraded speech is closer to the clean speech, which usually means better intelligibility. It can be calculated by the ratio of the power of the clean speech signal to the power of the difference between the clean and degraded signals:

$$SDR = 10 \log_{10} \left(\frac{(\mathbf{x}_{j,m}(n))^2}{(\bar{\mathbf{y}}_{j,m}(n) - \mathbf{x}_{j,m}(n))^2} \right) \geq \beta \quad (43)$$

Then, the intermediate intelligibility measure estimates the similarity between the clean and degraded temporal envelopes. To be more specific, it compares how similar the shapes of these envelopes are, indicating how well the important temporal features of the speech are preserved. It is referred to as $d_{j,m}$ and can be calculated as:

$$d_{j,m} = \frac{(\mathbf{x}_{j,m} - \mu_{\mathbf{x}_{j,m}})^T (\bar{\mathbf{y}}_{j,m} - \mu_{\bar{\mathbf{y}}_{j,m}})}{\|\mathbf{x}_{j,m} - \mu_{\mathbf{x}_{j,m}}\| \|\bar{\mathbf{y}}_{j,m} - \mu_{\bar{\mathbf{y}}_{j,m}}\|}, \quad (44)$$

where $\mu_{(\cdot)}$ refers to the sample average of the corresponding vector.

The final intelligibility score is an average of these intermediate measures over all frames and frequency bands, providing a comprehensive assessment of speech intelligibility. Finally, the average of the intermediate intelligibility measure over all bands and frames is calculated as:

$$d = \frac{1}{JM} \sum_{j,m} d_{j,m} \quad (45)$$

where M represents the total number of frames and J the number of one-third octave bands. So, the STOI measure provides an objective way to assess speech intelligibility by comparing clean and degraded speech signals. This score will be used to determine if the recovered signals have a certain quality and maintain high intelligibility for listeners, without needing to analyze them with the ear. To implement the STOI evaluation metric a MATLAB function is used.

5 Results

The results obtained by following the steps discussed in the report will now be presented. The estimation of the ATF was done in two ways by using the regular eigenvalue decomposition with pre-whitening from section 2.1, and by using the GEVD from section 2.2. Three beamformers discussed in section 3 are implemented and the evaluation metrics in terms of STOI and output SNR are shown for all of them. Lastly, six scenarios/experiments are developed to compare the performance of different ATF estimations and beamformer performances under different settings.

5.1 Eigenvalue Decomposition

Below, the performance of the beamformers DAS, MVDR, and Optimal Linear Multi-Channel Wiener (OLMCW) can be seen for the regular eigenvalue decomposition case.

Table 2: STOI scores for different scenarios and beamformers using the eigenvalue decomposition.

SCENARIOS	STOI		
	DAS	MVDR	OLMCW
Clean Speech 1 with Artificial Non-Stationary Noise	0.8341	0.8270	0.8280
Clean Speech 1 with Stationary Speech Shaped Noise	0.2603	0.7605	0.7624
Clean Speech 1 with Babble Noise	0.2208	0.7306	0.7255
Clean Speech 1 with All Noises	0.1773	0.6858	0.6826
Clean Speech 1 with Clean Speech 2	0.7337	0.7451	0.7340
Clean Speech 1 with All Noises and Clean Speech 2	0.1871	0.6667	0.6614

Table 3: Output SNR for different scenarios and beamformers using the eigenvalue decomposition.

SCENARIOS	SNR [dB]		
	DAS	MVDR	OLMCW
Clean Speech 1 with Artificial Non-Stationary Noise	30.5082	33.9497	33.9497
Clean Speech 1 with Stationary Speech Shaped Noise	9.4812	30.4410	30.4410
Clean Speech 1 with Babble Noise	10.6383	29.3826	29.3826
Clean Speech 1 with All Noises	2.3057	22.9352	22.9352
Clean Speech 1 with Clean Speech 2	32.1536	34.2415	34.2415
Clean Speech 1 with All Noises and Clean Speech 2	2.2153	22.8486	22.8486

5.2 Generalized Eigenvalue Decomposition

Below, the performance of the beamformers DAS, MVDR, and Optimal Linear Multi-Channel Wiener (OLMCW) can be seen for the GEVD case.

Table 4: STOI scores for different scenarios and beamformers using the generalized eigenvalue decomposition.

SCENARIOS	STOI		
	DAS	MVDR	OLMCW
Clean Speech 1 with Artificial Non-Stationary Noise	0.8063	0.7376	0.7374
Clean Speech 1 with Stationary Speech Shaped Noise	0.7739	0.7927	0.7913
Clean Speech 1 with Babble Noise	0.7324	0.7677	0.7666
Clean Speech 1 with All Noises	0.6412	0.7055	0.7050
Clean Speech 1 with Clean Speech 2	0.7449	0.6768	0.6763
Clean Speech 1 with All Noises and Clean Speech 2	0.6255	0.6886	0.6876

Table 5: Output SNR for different scenarios and beamformers using the generalized eigenvalue decomposition.

SCENARIOS	SNR [dB]		
	DAS	MVDR	OLMCW
Clean Speech 1 with Artificial Non-Stationary Noise	74.6632	80.7436	80.7436
Clean Speech 1 with Stationary Speech Shaped Noise	84.9184	84.9572	84.9572
Clean Speech 1 with Babble Noise	80.1511	80.3694	80.3694
Clean Speech 1 with All Noises	69.8427	70.2073	70.2073
Clean Speech 1 with Clean Speech 2	79.4307	81.7628	81.7628
Clean Speech 1 with All Noises and Clean Speech 2	70.1096	70.3905	70.3905

6 Discussion

In terms of the STOI metric, analyzing Tables 2 and 4 a clear distinction By first looking at the STOI metric results shown in Tables 2, 3, it is seen that the best results overall were achieved for the scenario with the stationary speech-shaped noise. This is expected because this noise had the lowest magnitude of all and is stationary. When testing the performance using the non-stationary noise source, it can be seen that the performance is significantly worse. The main difference between the two $\mathbf{R}_s$ methods can be seen in the results for the DAS beamformer. The EVD-based approach shows very changing results, which could be caused by the fact that this is the most simple beamformer. The beamformer can be called a ‘hit or miss beamformer’ for this approach. Looking at the scenarios more specifically the babble noise, shows a slight decrease in both intelligibility as STOI and quality as SNR can be deducted. This is a slightly harder scenario than the former two since babble noise represents crowded places where multiple conversations may overlap. However, when the audio output of this scenario is listened to, the degraded performance turns out to be not that significant and it can be said that the beamformers are still able to isolate the primary speech signal. Considering all noises, the effect of multiple sources with different characteristics is tested. The tables show that the interpretability is lower than the EVD-based result, which could

also be heard. When the scenario with the two clean speeches is analyzed, it is seen that for the MVDR and OLMCW, the STOI outcome is a bit behind those of the earlier scenarios. However, for the DAS the STOI is better than the previously mentioned cases. This can be heard in the final audio file as well. For all three beamformers, the outcomes are relatively good, especially for the first half of the audio file. During the first half, four female speakers are speaking for Clean Speech 2 and their contribution to the final result is very faint. Whereas for the second half, the four male speakers' voice begins to suppress the female voice in Clean Speech 1.

Combining every noise signal and the second signal as the interference sources presents the beamformer's hardest challenge. All different types of noises are combined with their own unique characteristic as mentioned earlier. The results obtained for the GEVD-based approach, are in line with the expectations because they are the lowest values seen. Hearing the obtained signals, the estimated clean speech is still distinguishable. For the EVD-based approach, this is only the case for the estimated speech using MVDR and the OLMCW. Overall, using the EVD to estimate the covariance matrix of the target signal gives more uneasy end results regarding sound quality. So, although the STOI values are relatively good, the pleasantness of hearing is not really taken into account.

In terms of the SNR metric, analyzing Tables 3 and 5 reveals a much lower SNR where EVD is used. This is particularly evident when the DAS beamformer is used, as it exhibits the poorest performance in the challenging scenario involving a combination of all noise types and two clean speech signals. For the case of GEVD, Table 5 presents consistent results, ranging between 70 to 85 dB.

Due to possible randomness in the behaviour of noise, the calculation of $\mathbf{R}_n$ and $\mathbf{R}_s$ is done as a weighted average of the previous calculated- and the new covariance matrix. This weighting is for this paper chosen as 0.9, meaning that the $\mathbf{R}$ is the addition of 90% past and 10% new calculated $\mathbf{R}$ one. This value has not been researched deeply, which could be a nice addition to future research. Something that could have been added to the evaluation of different methods presented to retrieve the signal was exploring other beamformers, such as LCMV. This beamformer can handle multiple constraints and thus can perform better when these constraints help mitigate noise sources.

7 Conclusion

To summarise, finding the covariance matrix of the target signal using the generalized eigenvalue decomposition is overall the more stable method. For the scenarios discussed in this paper, EVD works better for artificial non-stationary noise and clean speech. GEVD is more effective for stationary and babble noise, and combinations thereof. In the end, both methods of derivation should be evaluated for the specific use case, because both can give sufficient results and the superior method can differ. Regarding the beamformers, it showed that the more complex ones performed overall better with higher and more stable results for STOI and SNR.

References

- [1] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, “An algorithm for intelligibility prediction of time-frequency weighted noisy speech,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 19, no. 7, pp. 2125–2136, Sep. 2011.